

Education

University of Toronto - BAsC, Engineering Science (Major in **Robotics Engineering**) 2022 – 2027 (expected)
Minors: Engineering Business (Rotman) | **Relevant Courses:** Robotics, Mathematics for Robotics, Dynamics, Controls

Experience

Embedded Software Intern – Motor Control R&D Team - [Hedgehog Technologies](#) – Burnaby, BC May 2026 –

- Integrated user-defined parameter input functionality between Python GUI & controller firmware via CAN/Modbus, developing new data structures to send, receive and apply input to the control system, enabling real-time motor tuning.
- Developed embedded C firmware for a virtual EEPROM on the NXP S32K144 using internal flash peripheral drivers, implementing Hardware Abstraction Layers that enabled the team to remove the physical EEPROM from the design.
- Diagnosed repeated board faults to a MOSI-CLK short on the motor driver IC by probing the full SPI bus (MOSI, MISO, CS, CLK) and resolving via component replacement to restore nominal FOC operation.
- Identified and resolved bring-up failure modes (bootstrap capacitor sizing, failed MOSFETs) across 3 boards by tracing faults through Altium schematics; documented findings and component modifications to maintain accurate board status.

Process Engineering Intern – Process Optimization - [York Region](#) – Newmarket, ON May 2025 – April 2026

- Engineered a two-stage Python classifier (keyword filter + TF-IDF/Complement Naive Bayes) to triage 4,000–5,000 monthly operator logs, achieving 100% issue recall on validation data & reducing manual review volume by 55%.
- Derived geometric volume models for septage tanks from structural drawings, developing a linearized approximation in Excel suitable for SCADA integration and delivering live operational KPIs to $\pm 0.1 \text{ m}^3$ accuracy across pump stations.
- Built Python and Excel models from indirect operational data, informing EQ tank sizing decisions & saving \$18K annually.
- Drafted Python scripts to automate Excel-based data compliance graphs and tables cutting reporting time by 30%.

Selected Projects

Dynamic Stability Cart for Rope-Bridge Traversal ([Report](#)) Feb 2025 – Apr 2025

- Designed an autonomous balancing cart for rope bridge traversal using an STM32 Nucleo and 9-DoF IMU over SPI; cut control-loop latency by >95% via embedded C++ DMP sensor fusion and gain tuning on a PI-controlled servo rod.
- Validated stability through iterative hardware testing across dual- to single-anchor configurations, diagnosing Kd-induced oscillations and converging on a tuned PI controller ($K_p=1.5$, $K_i=0.02$) to achieve >80% crossing success rate.

Autonomous Mail Delivery Robot with Bayesian Localization ([GitHub](#)) Nov 2024 – Dec 2024

- Developed a ROS2 Jazzy autonomous mobile robot system using Python on Linux (Ubuntu), using a multi-threaded executor for concurrent perception & control nodes, integrating a PID line-following controller via camera feedback.
- Implemented a recursive Bayesian localization filter with a 12-state transition model ($P=0.85$ advance) & 5-class colour measurement model to resolve positional ambiguity across a closed-loop 12-state map, achieving >90% state confidence within 5 observations per trial; validated across 5 physical TurtleBot3 hardware trials, extended to Gazebo Harmonic.

Design Teams

Aerodynamics Lead - *University of Toronto Wind Turbine Team (UTWind)* Sept 2023 – June 2025

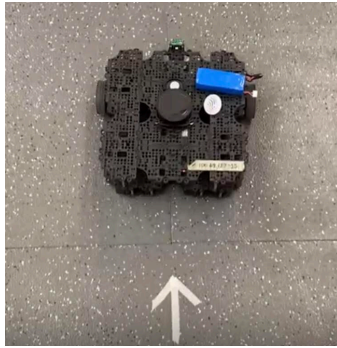
- Led a 10-member aerodynamics sub-team in iterative CFD optimization of VAWT blade profiles using ANSYS Fluent and Discovery, with results informing blade selection for 1st place at 2024 International Small Wind Turbine Competition.

Skills

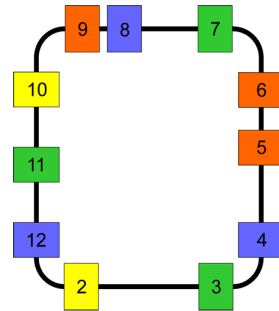
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- **Software:** C/C++, Object Oriented Programming, Python, Git/Github, Bash, Jira | **Protocols:** I2C, SPI, CAN, Modbus
 - **Robotics:** ROS2 (Jazzy), ROS1, Sensor Fusion, Autonomous Navigation, Multithreading, Embedded Systems
 - **Hardware:** Actuators/Servos, IMUs, Oscilloscope, Soldering, Signal Probing, Multimeter, Electrical Assembly
 - **Technologies/Tools:** ARM (STM32, NXP), Linux (Ubuntu ROS1/2), Command Line, Altium, Visual Studio, S32k IDE

Autonomous Mail Delivery Robot

[Final Report](#) | [Demo Video](#) | [Github](#)



TurtleBot3 Waffle Pi Autonomous Delivery Robot

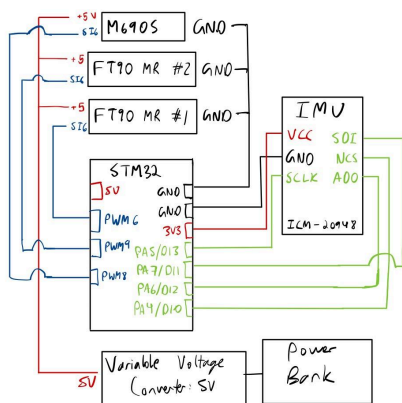


12-State Office Navigation Map

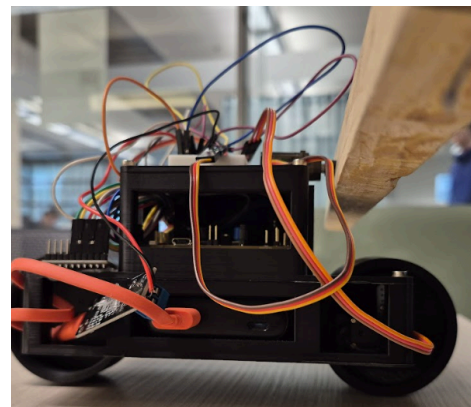
- Extended the physical **ROS1**/rospy TurtleBot3 implementation to **ROS2** Jazzy / **Gazebo** Harmonic, recalibrating colour classification thresholds from real camera to Ogre2 renderer while preserving the 12-state map and Bayesian filter.
- Applied a belief floor of 0.01 after every update cycle to prevent permanent state elimination, ensuring filter recovery from misclassified colour observations without irreversible state collapse.
- Implemented a corridor-freeze strategy suppressing filter prediction during mid-corridor traversal and firing exactly once on patch entry, preventing belief double-advancement between offices.
- Architected a two-phase state machine (exploration lap to delivery), gating delivery on 8 consecutive MAP-confident frames to prevent premature triggering; achieved 60% full 3-office delivery completion across 5 physical trials.

Dynamic Stability Cart for Rope-Bridge Traversal

[Final Report](#) | [Demo](#) | [Cart Overview](#)



System Diagram (STM32 & connections)



Side View of Cart with Balancing Beam

- Engineered a two-wheeled autonomous cart to traverse a custom-built rope-suspended bridge, integrating an **STM32** NUCLEO-G070RB with a 9-DoF ICM-20948 IMU over **SPI** and a servo-actuated balancing rod; developed embedded C++ firmware implementing PI control for real-time pitch stabilization, achieving $<3^\circ$ steady-state error and consistent successful bridge crossings on a 1 m single-axis rail.
- Reduced control-loop latency by $>95\%$ (5s to <100 ms) by offloading quaternion-based sensor fusion to the IMU's onboard DMP, eliminating tip-over events caused by delayed pitch correction.
- Developed two **Fusion 360** design iterations of 3D-printed chassis, transitioning from a high-wall enclosed structure to an open multi-level layout that reduced mass while maintaining structural integrity; enforced chassis symmetry to center the CoM on the cart's longitudinal axis, simplifying balancing problem and improving disturbance rejection on 1-axis rail.
- Validated system stability through iterative hardware testing across progressively harder rail configurations (dual-anchor to single-anchor string attachment), diagnosing Kd-induced oscillations that caused tip-overs and simplifying to a tuned PI controller ($K_p=1.5$, $K_i=0.02$) to achieve repeatable autonomous bridge traversal.